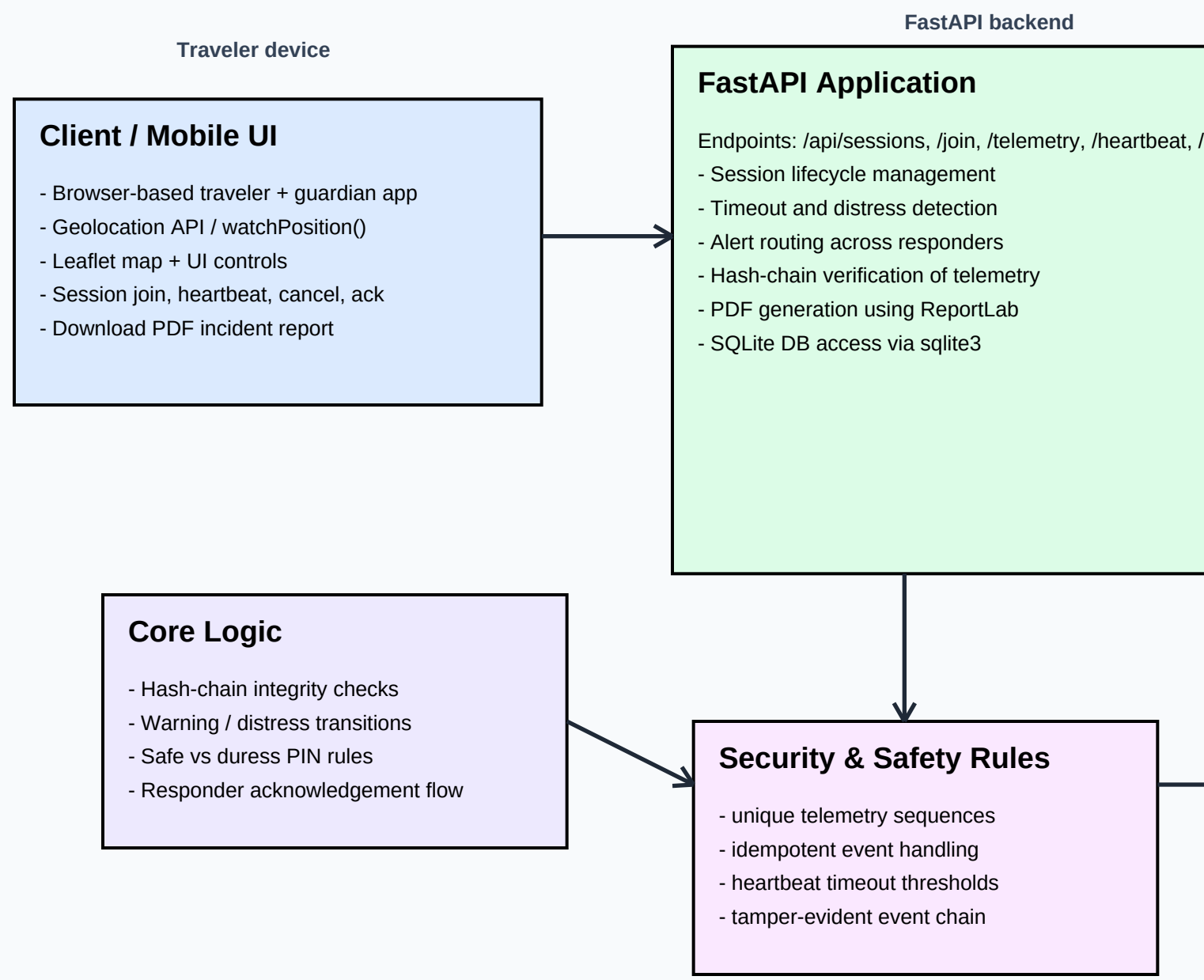


SentinelTrace Architecture Overview

Emergency escort tracking system for traveler safety, responder alerts, and incident reporting.



Tech Stack

Frontend:	HTML, CSS, JavaScript, Leaflet, browser geolocation
Backend:	Python 3, FastAPI, Pydantic, Uvicorn
Database:	SQLite with session, telemetry, alert, guardian and audit tables

How the System Works

- Reporting starts an escort session for the guardian and generates a session ID link.
- Guardian joins the escort and can share location while both devices monitor the route.
- Browser GPS telemetry is sent to FastAPI with sequence numbers and timestamps.
- FastAPI validates sessions, stores telemetry, creates a hash chain, and updates state.
- If heartbeats stop, the system transitions from ACTIVE to WARNING to DISTRESS and creates alerts.